

Computer Networks Spring 2026 Assignment#4 (6A & 6B) Solution

Question 1:

Total Size: 1060 bytes.

- Original Payload: 1000 bytes
- IPv6 Header: 40 bytes (Total IPv6 Datagram = 1040 bytes)
- IPv4 Header (Encapsulation): 20 bytes
- **Calculation:** 1040 (IPv6 packet) + 20 (IPv4 header) = 1060 bytes. The entire IPv6 packet becomes the *payload* of the IPv4 packet.

Question 2:

NAT violates the principle because routers are supposed to process packets only up to the Network Layer (Layer 3). NAT modifies fields (IP address, Port numbers) that are technically Transport Layer (Layer 4) information. It prevents hosts from being uniquely addressable from the outside world without state maintained in the middlebox.

Question 3:

- **Reason:** The NAT router's translation table has no entry for an *incoming* unsolicited connection. It doesn't know which internal IP to forward the traffic to.
- **Fix: Port Forwarding** (or UPnP). You must statically map a port on the router (e.g., 6881) to the specific internal IP of the P2P client.

Question 4:

- **Checksum:** Removed entirely (relies on Layer 2 and Layer 4 checksums).
- **Fragmentation (ID/Flags/Offset):** Moved to Extension Headers (routers no longer frag/defrag packets).
- *Also acceptable:* Header Length (IHL) is removed because the base header is fixed at 40 bytes.

Question 5:

- | | | |
|-----------------------------|--|----------------------|
| A. Match: Src.IP= 10.1.0.1, | Dest.IP = 10.2.0.4 | Action: Drop |
| B. Src.IP= 10.1.*.*, | Dest.IP= 10.3.0.5, Dest.Port= 80 OR 25 | Action: Forward at 1 |
| C. Src.IP= 10.1.*.*, | Dest.IP= 10.3.0.5, Dest.Port != 80 | Action: Drop |
| D. S3->S2->S1 | | |
| • At S3: | | |
| Match: Src.IP= 10.3.0.5, | Dest.IP= 10.1.0.1 | Action: Forward at 4 |
| • At S2: | | |
| Match: Src.IP= 10.3.0.5, | Dest.IP= 10.1.0.1 | Action: Forward at 2 |
| • At S1: | | |
| Match: Src.IP= 10.3.0.5, | Dest.IP= 10.1.0.1 | Action: Forward at 2 |

Question 6:

Router R1 decides after consulting its forwarding table that the original datagram needs to go to router R2. But the link from R1 to R2 has MTU of 1500 Bytes while the original datagram is of size 1600 Bytes. Router looks at the DF flag (whose value is 0) to verify that it is allowed to do IPv4 datagram fragmentation.

Payload size = $1600 - 20 = 1580$ Bytes

First fragment will have = 20 Bytes header + 1480 Bytes of payload

Second fragment will have = 20 Byte IPv4 header + 100 Bytes of payload

(Recall that fragment offset count blocks of 8 Bytes)

Total Length (Bytes)	IP Header Length (Bytes)	Identification (ID) (Hexadecimal)	DF Flag	MF Flag	Fragment Offset
1500	20	0xCAFE	0	1	0
120	20	0xCAFE	0	0	185

Since there is a single path from R1 to R2, no packet re-ordering will happen and above fragments will arrive in order at router R2.

MTU from R2 to Host B = 512 Bytes

$512 - 20 = 492$ Bytes payload possible

But 8 does not evenly divide 492 (Offset counts in blocks of 8 Bytes)

So, we can send max payload of size 488 Bytes in each datagram from R2 to B.

Total Length (Bytes)	IP Header Length (Bytes)	Identification (ID) (Hexadecimal)	DF Flag	MF Flag	Fragment Offset
508	20	0xCAFE	0	1	0
508	20	0xCAFE	0	1	61
508	20	0xCAFE	0	1	122
36	20	0xCAFE	0	1	183

Second fragment coming from R1 does not need further sub-fragmentation:

Total Length (Bytes)	IP Header Length (Bytes)	Identification (ID) (Hexadecimal)	DF Flag	MF Flag	Fragment Offset
120	20	0xCAFE	0	0	185

Question 7:

Department /Office	Subnet Mask	Network Address	First Usable Host Address	Broadcast Address	No. of free addresses
CS	255.255.255.128/25	192.168.48.0	192.168.48.1	192.168.48.127	27
IT	255.255.255.192/26	192.168.48.128	192.168.48.129	192.168.48.191	0
Finance	255.255.255.192/26	192.168.48.192	192.168.48.193	192.168.48.255	12
Admin	255.255.255.224/27	192.168.49.0	192.168.49.1	192.168.49.31	10
Director	255.255.255.240/28	192.168.49.32	192.168.49.33	192.168.49.47	5
No. of Free addresses	720 addresses (192.168.49.48 to 192.168.51.255)				818

Used address space from 192.168.48.0 to 192.168.49.47

Remaining free addresses inside the /22 address space: 192.168.49.48 to 192.168.51.255

Which means 720 addresses are still left) i.e., $208+256+256 = 720$ are left while 304 addresses are allocated.

Category	Count
Total IPs in /22	1024
Allocated subnets	304 IPs (raw block size)
Unallocated block	720 IPs
→ Usable in unallocated block	718 IPs (excluding 1 network + 1 broadcast)
→ Usable in allocated subnets	294 IPs
Total usable IPs overall	$294 + 718 = 1012$ ✓
Usable IPs still available	818 ✓

So What Is “818 Usable Leftover”?

It’s this:

- Total usable IPs = 1012
- Already assigned to departments = 294
- Still available for future use = $1012-294=818$

 Final Summary		
Metric	Value	Meaning
Total IPs in /22	1024	All addresses
Total usable host IPs	1012	Excludes all network/broadcast
Usable IPs already assigned	294	Inside CS, IT, Finance, Admin, Director
Usable IPs still available	818	What’s left for future subnets

Solution: (Reverse Allocation)

Department	Prefix	Subnet Mask	Network Address	First Usable Host	Broadcast Address
CS	/25	255.255.255.128	192.168.51.128	192.168.51.129	192.168.51.255
IT	/26	255.255.255.192	192.168.51.64	192.168.51.65	192.168.51.127
Finance	/26	255.255.255.192	192.168.51.0	192.168.51.1	192.168.51.63
Admin	/27	255.255.255.224	192.168.50.224	192.168.50.225	192.168.50.255
Director	/28	255.255.255.240	192.168.50.240	192.168.50.241	192.168.50.255

Solution: (Middle Allocation)

Question 8:

- (i) Maximum number of hosts in the parent IP address block: $2^9 - 2 = 512 - 2 = 510$
- (ii) Subnet Address of Subnet A (in CIDR notation): **105.83.211.128/27**
- (iii) First assignable IP in Subnet A (in CIDR notation): **105.83.211.129/27**
- (iv) Broadcast address of Subnet A (in CIDR notation): **105.83.211.159/27**
- (v) Subnet Address of Subnet B (in CIDR notation): **105.83.210.0/24**
- (vi) First assignable IP in Subnet B (in CIDR notation): **105.83.210.1/24**
- (vii) Last assignable address of Subnet B (in CIDR notation): **105.83.210.254/24**
- (viii) Subnet Address of Subnet C (in CIDR notation): **105.83.211.0/25**
- (ix) Last assignable IP in Subnet C (in CIDR notation): **105.83.211.126/25**
- (x) Broadcast address of Subnet C (in CIDR notation): **105.83.211.127/25**

Question 9:

- I. Write the subnet mask for each group of customers.

Answer:

Subnet mask for first group customers: **255.255.255.0**

Subnet mask for third group customers: **255.255.255.240**

- II. Write the network address for 1st customer of each group:

Answer:

Network address of 1st customer of first group: **170.25.96.0/24**

Network address of 1st customer of second group: 170.25.104.0/26

Network address of 1st customer of last group: 170.25.108.0/28

III. Write the broadcast address of last customer of each group:

Answer:

broadcast address of last customer of first group: 170.25.103.255/24

broadcast address of last customer of second group: 170.25.107.255/26

broadcast address of last customer of third group: 170.25.111.255/28